DhakshaGroundControl Camera & Streaming SOP

Oct 24, 2026 · @mariappan chellam

About this manual

This manual explains, step by step and in plain words, how to use the drone camera and live video streaming in DhakshaGroundControl, the ground control app on the MK15 / RC controller.

- **Who it is for:** pilots and camera operators in the field, no technical knowledge needed.
- **What it covers:** the Skydroid day and night camera, the SIYI ZR10 camera, the streaming settings, and how the live video reaches the office (DhakshaKey, the Dhaksha video server and Dhaksha Live).
- **App version:** DhakshaGroundControl 4.2 build of 24 October 2026 (it includes the Night palette setting).

Words you will see:

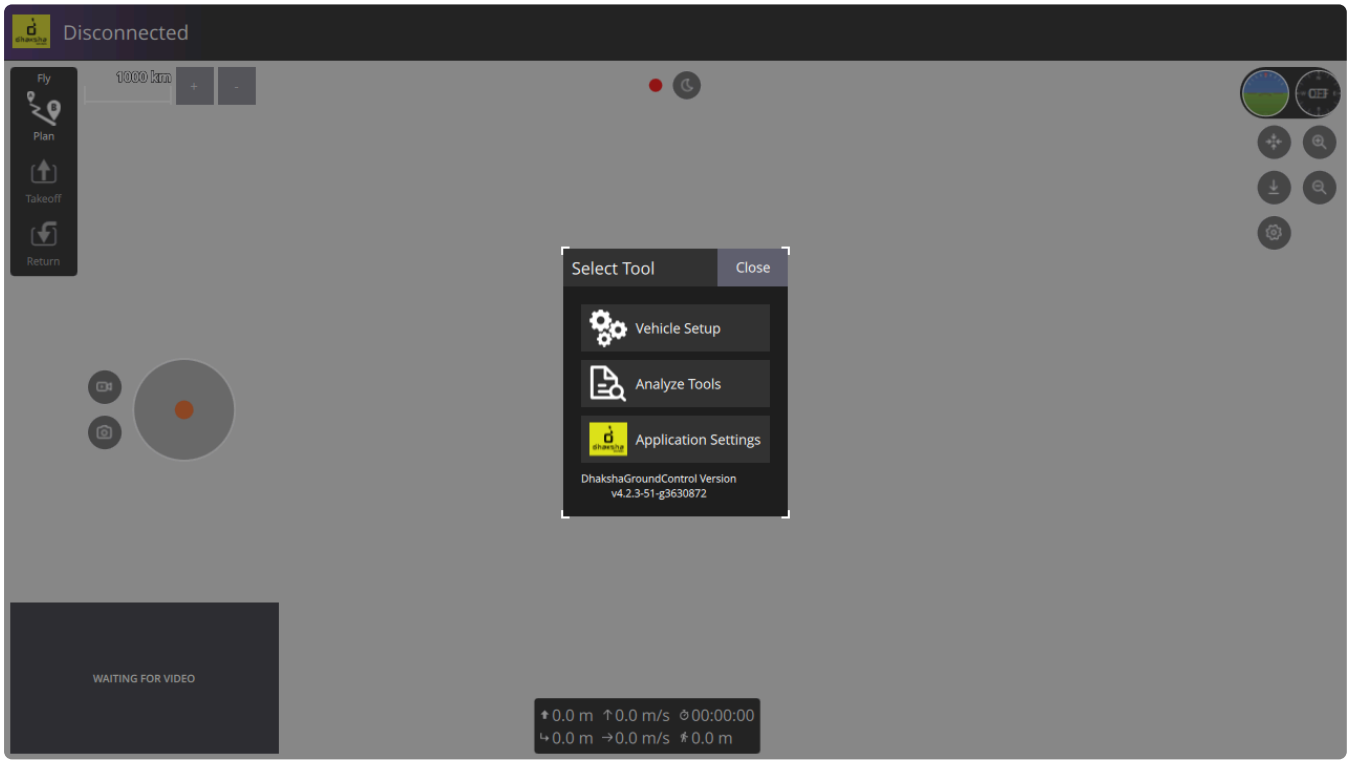
Word	Meaning
Gimbal	The motorised holder that turns and tilts the camera
Day camera	The normal colour camera
Night camera	The thermal (heat) camera; it sees warm objects in the dark
PiP (picture-in-picture)	A small second picture in the corner of the screen
Stream	Live video sent over the internet
Server	The Dhaksha computer on the internet (15.252.170.151) that receives the drones' video and passes it on to viewers

1. One-time setup: tell the app which camera is fitted

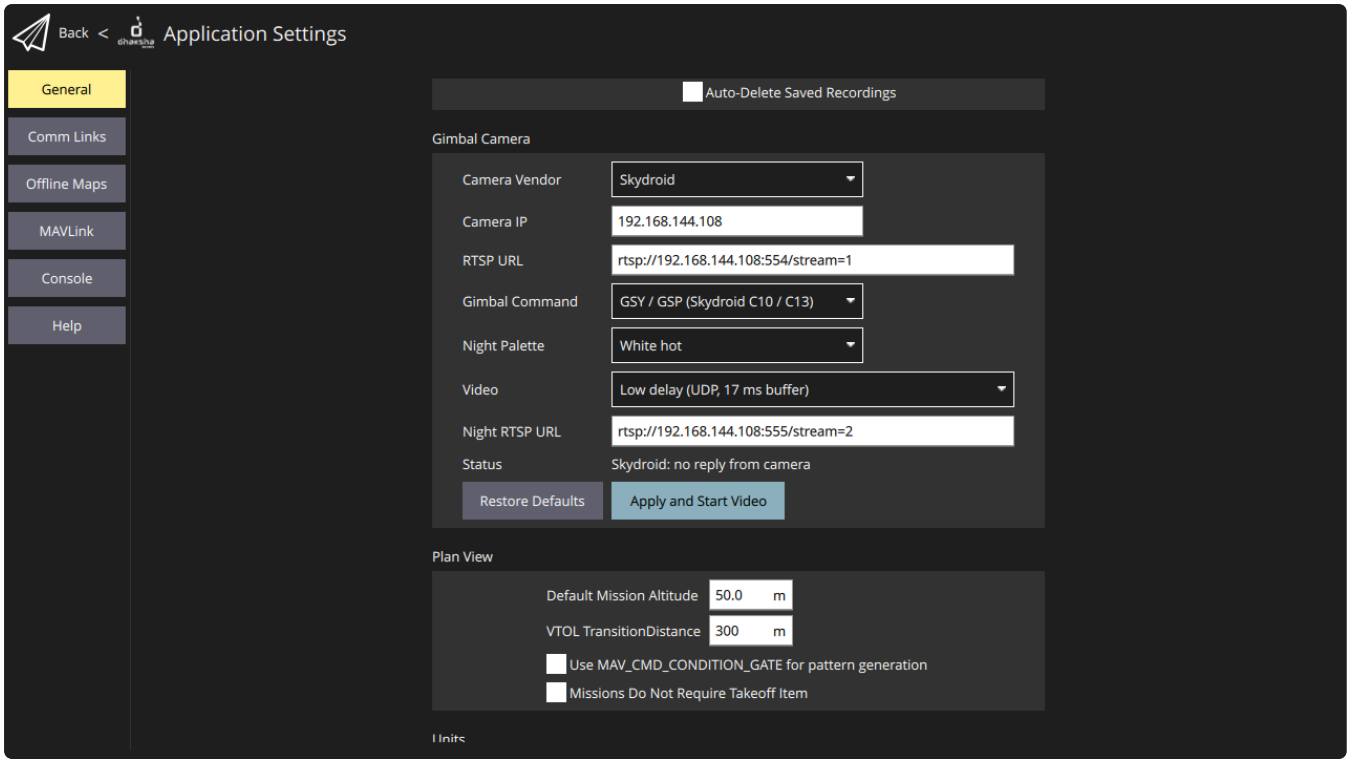
Do this once per drone, and again only if the camera is changed. The app remembers it.

1. Open DhakshaGroundControl.

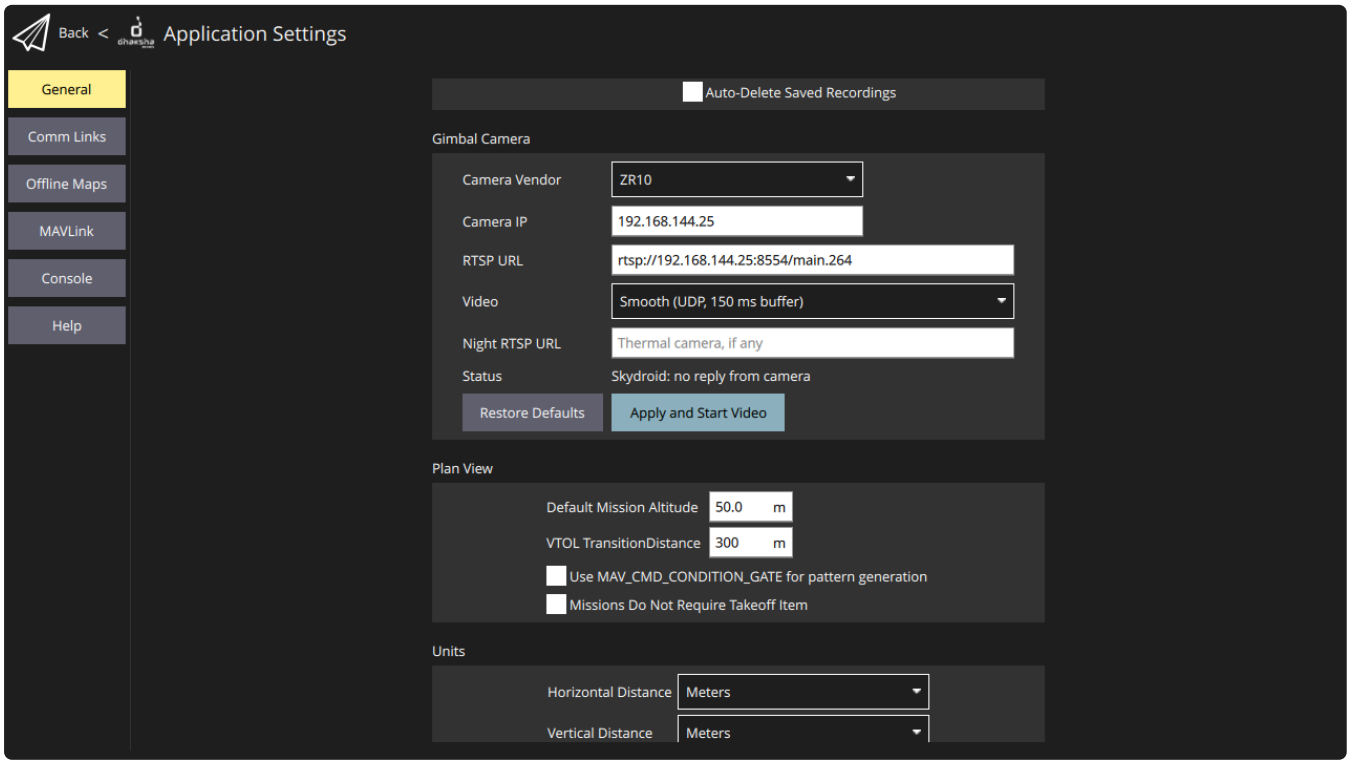
2. Tap the Dhaksha logo at the top left, then **Application Settings**, then **General**.
3. Scroll down to the **Gimbal Camera** box.
4. At **Camera Vendor**, choose **Skydroid** (day and night camera) or **ZR10**.
5. Tap **Restore Defaults**. This fills in the correct camera IP and video addresses.
6. At **Video**, choose how the picture is received (see the table below). Start with the suggested one.
7. Tap **Apply and Start Video**. The camera picture appears on the main screen.
8. Check **Status** in the same box: it should say **Skydroid: connected** or **ZR10: connected**.



Step 2: tap the Dhaksha logo at the top left, then Application Settings. The version is shown under it.



Steps 3-8 with Skydroid: Camera Vendor, the filled-in addresses, Gimbal Command, Night Palette, Video, Restore Defaults and Apply and Start Video.



The same box with ZR10: no Gimbal Command or Night Palette; Video set to Smooth.

Default values filled in by Restore Defaults:

Setting	Skydroid	ZR10
Camera IP	192.168.144.108	192.168.144.25
RTSP URL (day video)	rtsp://192.168.144.108:554/stream=1	rtsp://192.168.144.25:8554/main.264
Night RTSP URL	rtsp://192.168.144.108:555/stream=2	(empty, no night camera)
Gimbal Command	GSY / GSP (Skydroid C10 / C13)	not shown
Night Palette	White hot (change any time)	not shown

Video setting, which to choose:

Video choice	What it does	Suggested for
Low delay (UDP, 17 ms buffer)	Picture is almost instant; may break up if the radio link is weak	Skydroid
Smooth (UDP, 150 ms buffer)	About 0.15 s later, but fewer freezes	ZR10
Reliable (TCP)	Never breaks up, but delay can grow on a weak link	Only if the other two freeze a lot

If **Status** says **no reply from camera**: check the drone is powered on, the radio link is connected, and the Camera IP matches the table. Then tap **Apply and Start Video** again.

2. Skydroid day and night camera

The Skydroid camera has two cameras in one: a **day** (colour) camera and a **night** (thermal) camera. You control both from round buttons on the flight screen.

Where the buttons are



1 Connection dot · 2 Day / Night · 3 Record · 4 Photo · 5 Joystick · 6 Centre · 7 Zoom in · 8 Tilt down · 9 Zoom out · 10 Gear · 11 Second camera box. (Screenshot without a drone connected, so the dot is red and there is no video.)

Place on screen	Button	What it does
Top centre	Connection dot	Green when the camera answers the app
Top centre	Day / Night	Switches the main picture between the day and the night camera
Left	Record	Starts recording on the camera's SD card; tap again to stop
Left	Photo	Takes one photo on the SD card
Left, next to Record and Photo	Joystick	Drag the knob to turn and tilt the camera
Right, under the HUD	Centre	Points the camera straight ahead
Right	Tilt down	Points the camera straight down (90°)
Right	Zoom in / Zoom out	Hold to zoom; let go to stop
Right	Gear	Gimbal speed and Night palette

Place on screen	Button	What it does
Bottom-left box	Second camera	Shows the camera that is not on the main screen

A short message under the top buttons confirms each action, for example "Gimbal centre" or "Night camera".

Before take-off

1. Check the connection dot is **green**.
2. Check there is no red **No SD card** or **SD card full** message. If there is, insert or empty the SD card before flying, or photos and videos will not be saved.
3. Tap **Centre** and check the camera moves.

Turning and tilting the camera

1. Put your thumb on the joystick knob and drag it gently in the direction you want the camera to look.
2. A small drag moves slowly for fine aiming; dragging further moves faster.
3. Let go to stop. The camera stays where it is.
4. Tap **Centre** to look straight ahead again, or **Tilt down** to look straight down.

To change how fast the camera can turn: tap the **Gear**, move the **Gimbal speed** slider (10 to 60 degrees per second), then tap **Confirm**. A lower speed makes aiming easier at high zoom.

Zoom

- Hold **Zoom in** or **Zoom out**. The level shows on screen (for example "Zoom 6x").
- The camera zooms itself up to 4x; beyond that the app enlarges the picture on screen, up to 30x in total. Above 4x the picture is less sharp because it is enlarged.

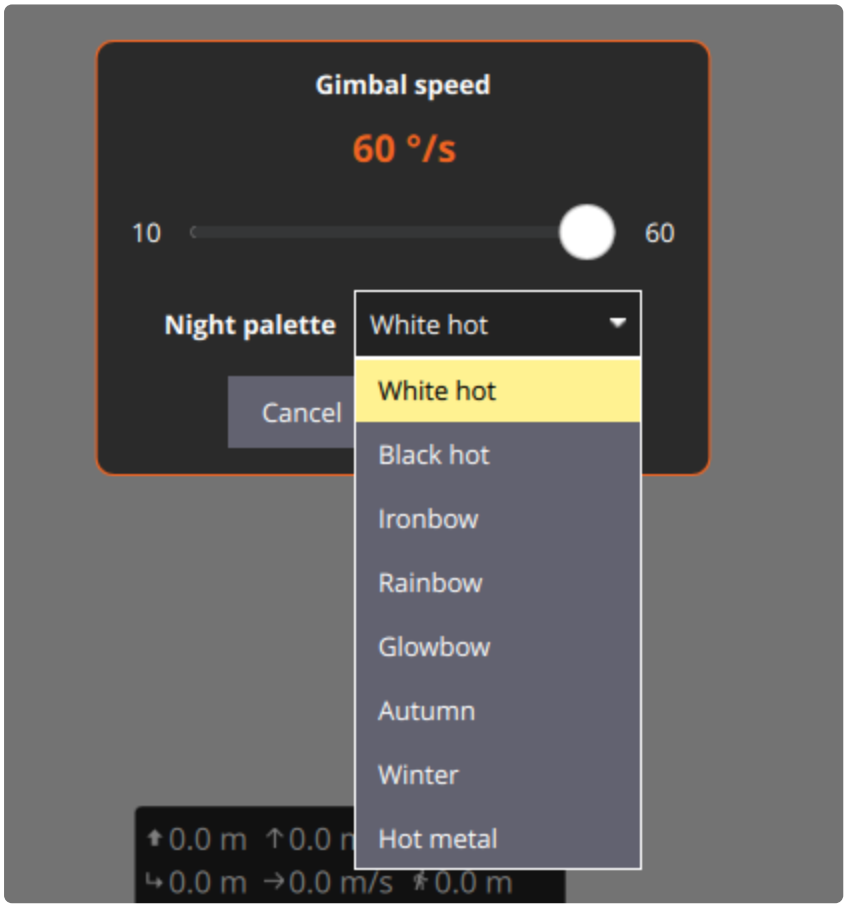
Day and night

- Tap **Day / Night** at the top to swap the main picture between the day and night camera.
- The **bottom-left box** shows the other camera at the same time. Tap the box to swap the two pictures.
- To see the map in the bottom-left box instead, tap the small **map** button on the box. To bring the camera back, tap the **camera** button on the map.

Night palette (colours of the night camera)

The night camera shows heat. The palette decides how heat is coloured.

1. Switch the main picture to the **night** camera.
2. Tap the **Gear**.
3. At **Night palette**, pick one. The picture changes straight away and the app shows, for example, "Night palette: Ironbow". You do not need to press Confirm.
4. The choice is saved and sent to the camera again every time it reconnects.



The Gear panel: Gimbal speed slider, and the Night palette list opened.

Palette	Looks like	Good for
White hot	Warm things white, cold things black	General use; people and vehicles at night
Black hot	Warm things black, cold things white	Same as White hot, some find it easier to read
Ironbow	Warm yellow-orange, cool purple-blue	Spotting hot spots (fires, engines)
Rainbow	Many colours	Seeing small temperature differences

Palette	Looks like	Good for
Glowbow, Autumn, Winter, Hot metal	Other colour styles	Try and keep the one that is clearest for the task

The palette setting is new in this build and has not yet been field-tested on every camera. If a palette does not change the picture, note which one and report it.

The same setting is also in **Application Settings > General > Gimbal Camera > Night Palette**.

Photos and videos

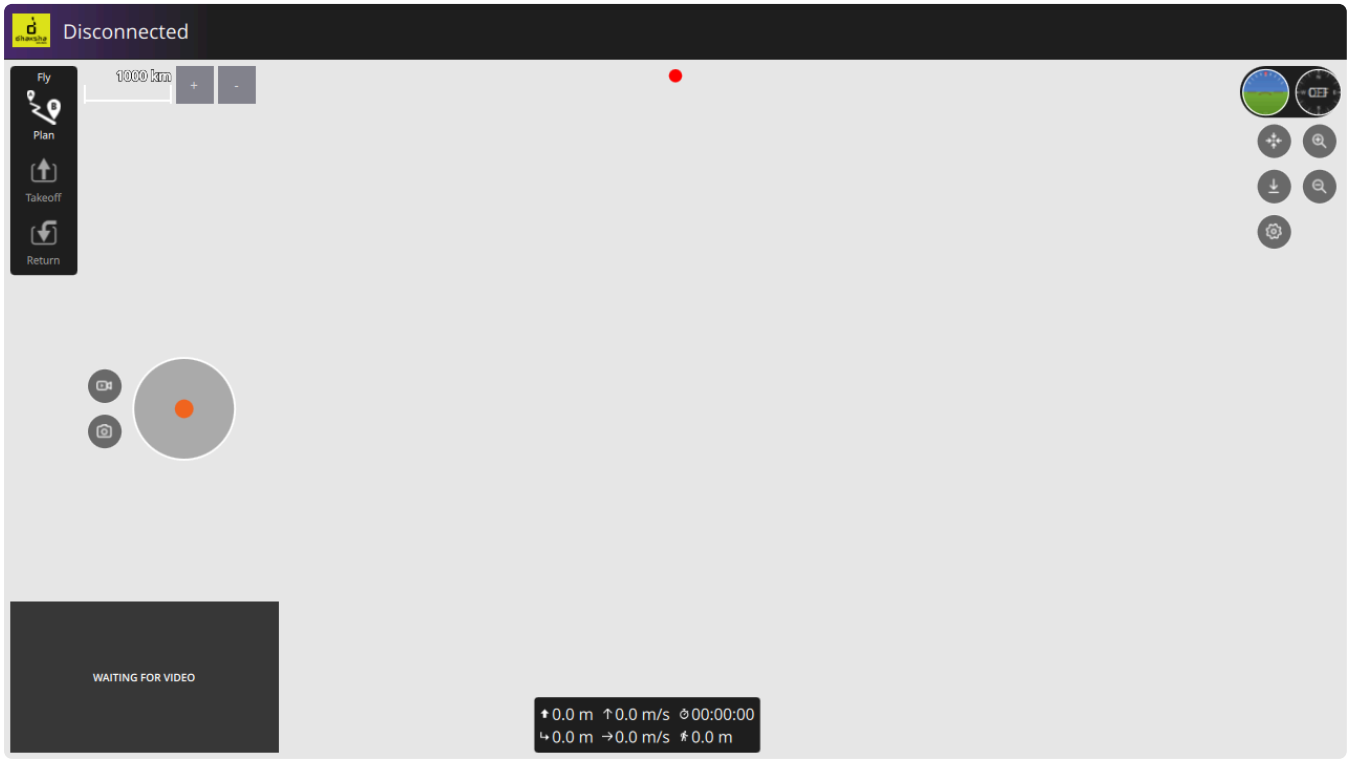
1. Tap **Record** to start. ● **REC** and the running time show at the top.
2. Tap **Record** again to stop. The message **Recording saved** appears.
3. Tap **Photo** to take a picture.
4. Photos and videos are stored on the camera's SD card, not on the controller.

3. SIYI ZR10 camera

The ZR10 is a day camera with strong zoom. It has no night camera, so there is no Day / Night button, no Night palette and no second camera in the bottom-left box (that box shows the map).

The app talks to the ZR10 directly over the drone's network cable link, the same way the SIYI FPV app does. Nothing needs to be changed in the autopilot.

Buttons



With ZR10 there is only the connection dot at the top: no Day / Night button.

The buttons are in the same places as for the Skydroid camera:

Place on screen	Button	What it does
Top centre	Connection dot	Green when the ZR10 answers
Left	Record	Start / stop video recording on the ZR10's SD card
Left	Photo	Take one photo on the SD card
Left	Joystick	Drag to turn and tilt the camera; further = faster
Right	Centre	Look straight ahead
Right	Tilt down	Look straight down (90°)
Right	Zoom in / Zoom out	Hold to zoom; the level shows, e.g. "Zoom 12.0x"
Right	Gear	Gimbal speed (10 to 60 degrees per second)

Step by step

1. Make sure **Camera Vendor** is **ZR10** in Application Settings (section 1) and the connection dot is green.

2. Before take-off, check there is no **No SD card** / **SD card full** message, and tap **Centre** to check the camera moves.
3. Aim with the joystick; let go to stop.
4. Hold **Zoom in** to zoom; let go to stop. Use a low **Gimbal speed** (Gear) when zoomed in far, so small moves are easy.
5. Tap **Photo** for a picture. The message **Photo taken** confirms it; **Photo failed** means it was not saved.
6. Tap **Record** to start and again to stop. If you see **Recording did not start**, tap Record again and check the SD card.

If the ZR10 video freezes or lags

- In Application Settings > General > Gimbal Camera, set **Video** to **Smooth (UDP, 150 ms buffer)**.
- If it still freezes often, try **Reliable (TCP)**.
- Tap **Apply and Start Video** after changing it.

4. Streaming settings inside DhakshaGroundControl

DhakshaGroundControl itself does not send video to the internet in this build. The **DhakshaKey** app on the same controller does that (section 5). In DhakshaGroundControl you only check two settings, so that the video and the drone's position reach DhakshaKey.

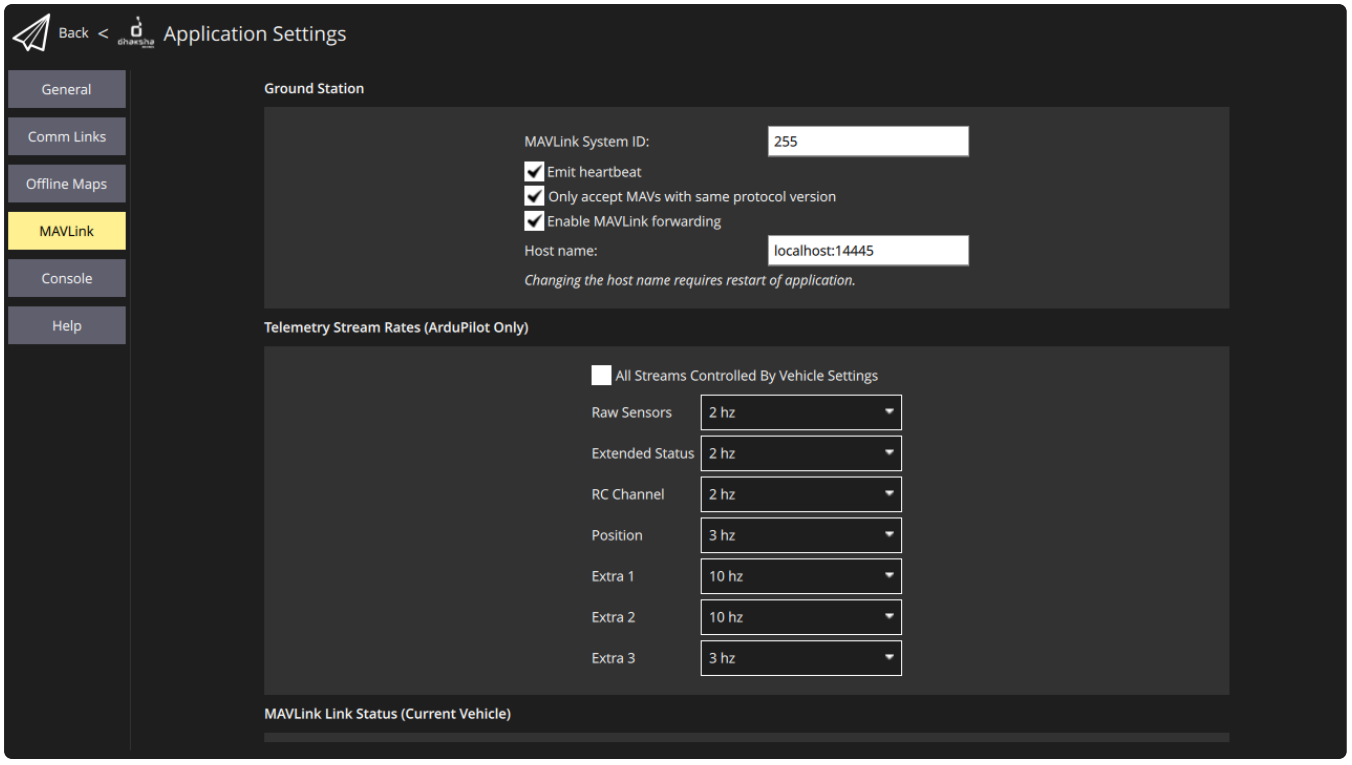
Setting 1: Video (how the controller receives the camera)

Application Settings > General > Gimbal Camera > **Video**. Choose as in section 1: **Low delay** for Skydroid, **Smooth** for ZR10, **Reliable** only if the picture keeps freezing. Tap **Apply and Start Video** after any change.

Setting 2: MAVLink forwarding (puts the drone's position on the streamed video)

DhakshaKey writes the drone's latitude, longitude and height at the bottom of the video sent to the office. It gets them from DhakshaGroundControl:

1. Application Settings > **MAVLink**.
2. Tick **Enable MAVLink forwarding**.
3. Host name: **localhost:14445**.
4. Restart DhakshaGroundControl once after changing this.



Application Settings > MAVLink: Enable MAVLink forwarding ticked, Host name localhost:14445.

If this is off, the video still streams, but it shows "waiting for position..." instead of the position.

Not used in this build

The **Stream to Server** option seen in older test builds is hidden. Always stream with DhakshaKey.

5. Live streaming to the office

Live video travels in four steps:

1. **Drone camera** (Skydroid or ZR10) sends video to the controller over the radio link.
2. **DhakshaKey** on the controller shrinks the video, adds the drone's position, and uploads it over the SIM or Wi-Fi internet.
3. The **Dhaksha video server** (15.252.170.151) receives every drone's video.
4. **Dhaksha Live** shows up to 25 drones at once, on a Windows or Linux PC, an Android phone, or any web browser.

5.1 DhakshaKey quick start (full guide in section 6)

First time only: install Termux and do the steps under **Setup and more** in DhakshaKey.

1. Open **DhakshaKey**.

2. **Camera**: choose Dhaksha - Day, Dhaksha - Night, Dhaksha - Day + Night (both at once) or Dhaksha - ZR10.
3. **Video quality**: start with **Good - 360p**. Use a lower one if the internet is weak.
4. Check **Camera IP** and **Server IP** (normally already filled in). If you change one, tap its **Save**.
5. **Drone**: choose this drone's number (1 to 25). The **Stream name** fills in as drone1, drone2... and drone1-night for the night camera. Keep these names: Dhaksha Live looks for them.
6. **Upload with**: leave on **RTMP** unless told otherwise (see the table below).
7. **Watch with**: HLS or WebRTC; this only changes the "Watch at" link shown.
8. Tap **Start streaming**. A Termux window opens and shows progress. The coloured bar at the top tells you how it is going.
9. Tap **Stop streaming** when done.

Video quality choices:

Quality	Picture	Internet needed (about)
Normal - 240p	426 x 240, 10 frames/s	0.1 Mbit/s
Good - 360p	640 x 360, 15 frames/s	0.15 Mbit/s
Very Good - 480p	854 x 480, 15 frames/s	0.3 Mbit/s
Exceptional - 720p	1280 x 720, 20 frames/s	0.8 Mbit/s

With Day + Night, two videos are sent, so the internet needed doubles.

Messages in the DhakshaKey bar:

Message	Meaning	What to do
Live: drone1 is on the server (green)	Working	Nothing
Data rate too low to stream video. Reduce resolution	Upload internet too slow	Choose a lower Video quality and Start again
No video from the camera	Camera not sending	Check drone power, radio link and Camera IP
Cannot reach the server	No internet, or wrong server IP / port	Check SIM / Wi-Fi, Server IP and Upload port

Message	Meaning	What to do
The stream dropped. Reconnecting… (orange)	Short break	Wait; it reconnects by itself
This phone's ffmpeg cannot upload with …	The chosen upload type is not available	Choose RTMP

5.2 Upload types ("Upload with")

Upload with	Port	When to use
RTMP	1935	Normal choice; works on most mobile networks
RTSP	8554	If RTMP is blocked on a network
SRT	8890	Weak, unsteady mobile networks; the server's firewall must allow port 8890 (UDP) first
WebRTC (WHIP)	8889	Lowest delay; needs a newer ffmpeg in Termux, otherwise DhakshaKey says it cannot be used

Whatever upload is used, viewers can watch over HLS or WebRTC.

5.3 Dhaksha Live (watching in the office or on a phone)

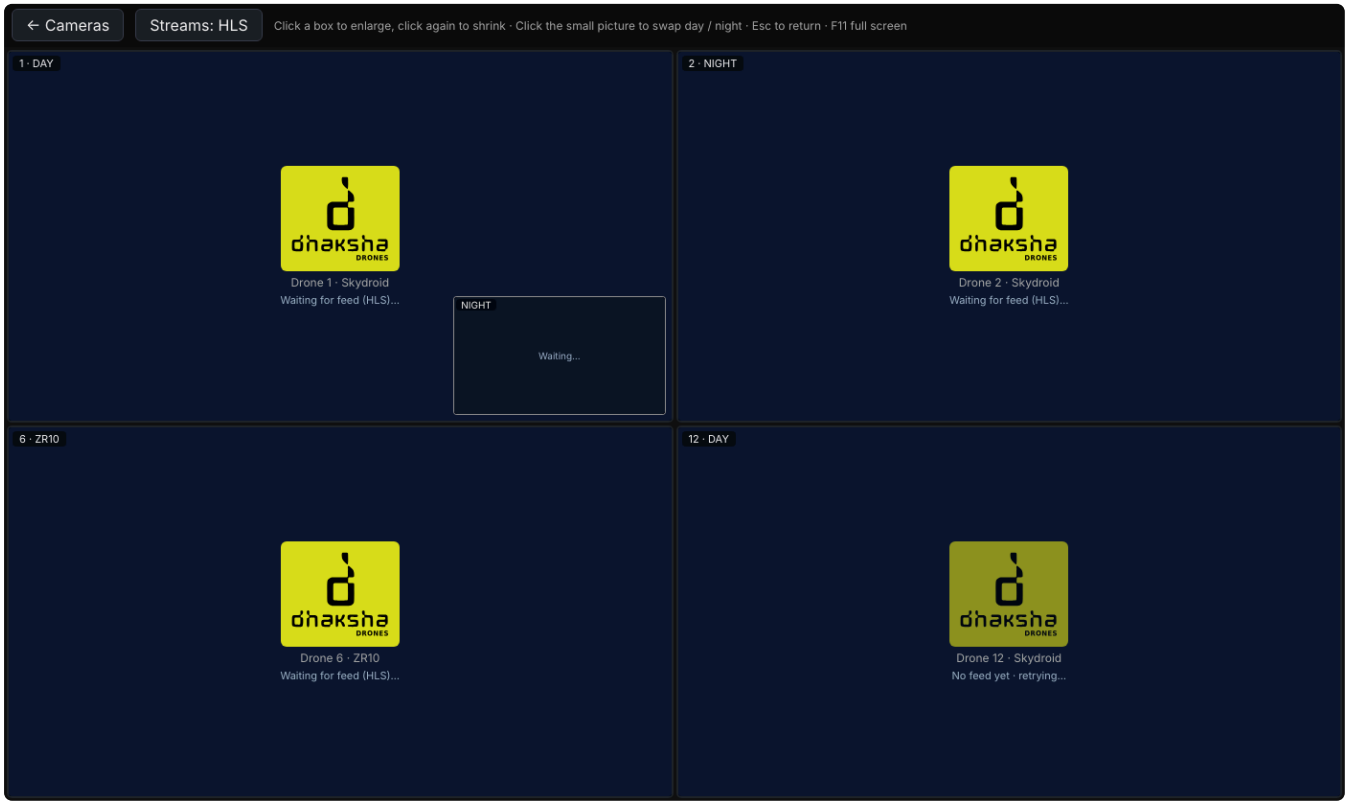
Ways to open it:

Device	How
Windows PC	Install the Dhaksha Live Windows app
Linux PC	Install the .deb, or run the AppImage / .tar.gz
Android phone	Install the Dhaksha Live app
Any browser	Open http://15.252.170.151 (or :8080)

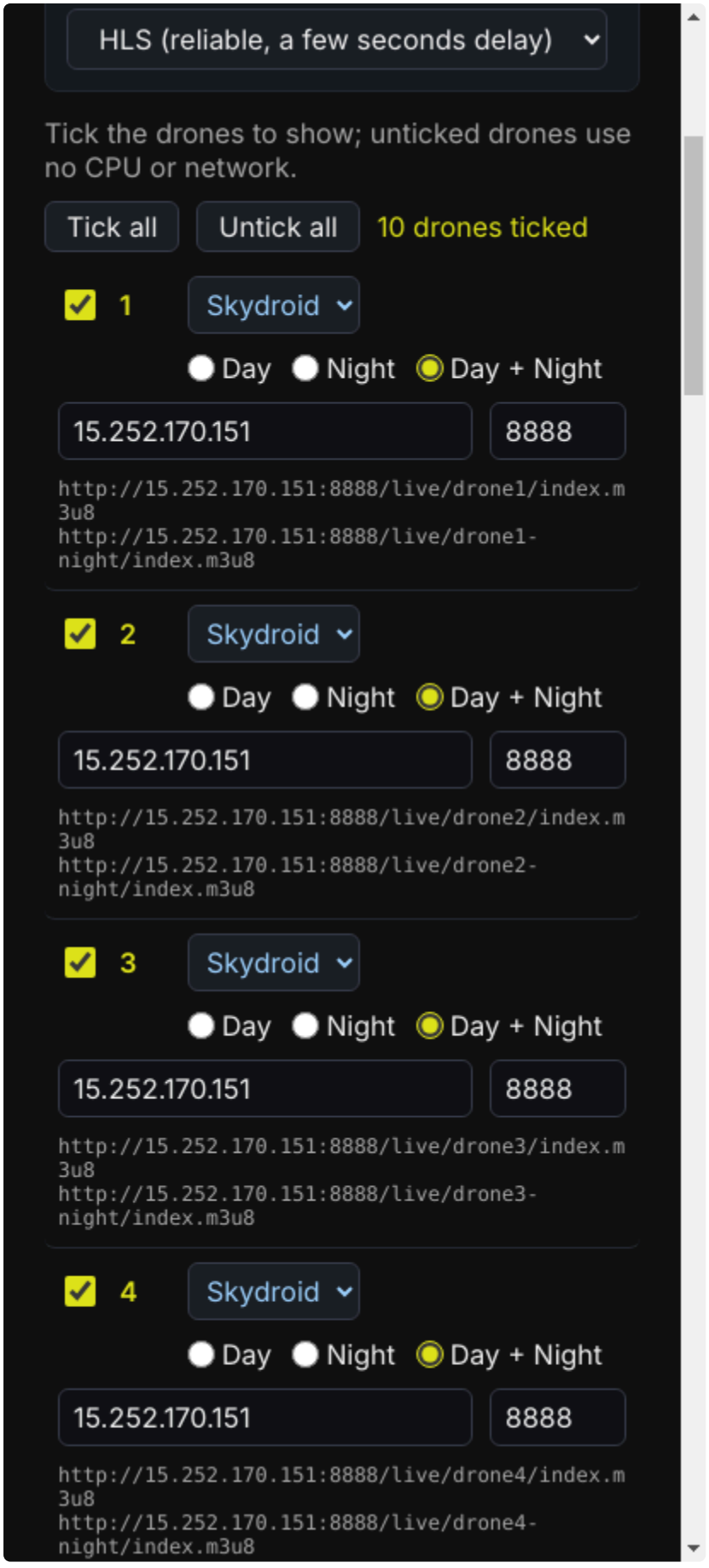
First time (camera window):

1. Enter **Server IP** 15.252.170.151 and **Port** 8888 (in a browser on a restricted network, use port 80), then tap **Use for all drones**.
2. For each drone, choose **Camera**: Skydroid or ZR10.
3. Choose **View**: for Skydroid, **Day**, **Night** or **Day + Night**; for ZR10 only **Day**.
4. Tick the drones to show (unticked drones use no internet).

5. Each row lists its stream address(es), for checking.
6. Tap **OK: show the ticked drones**.



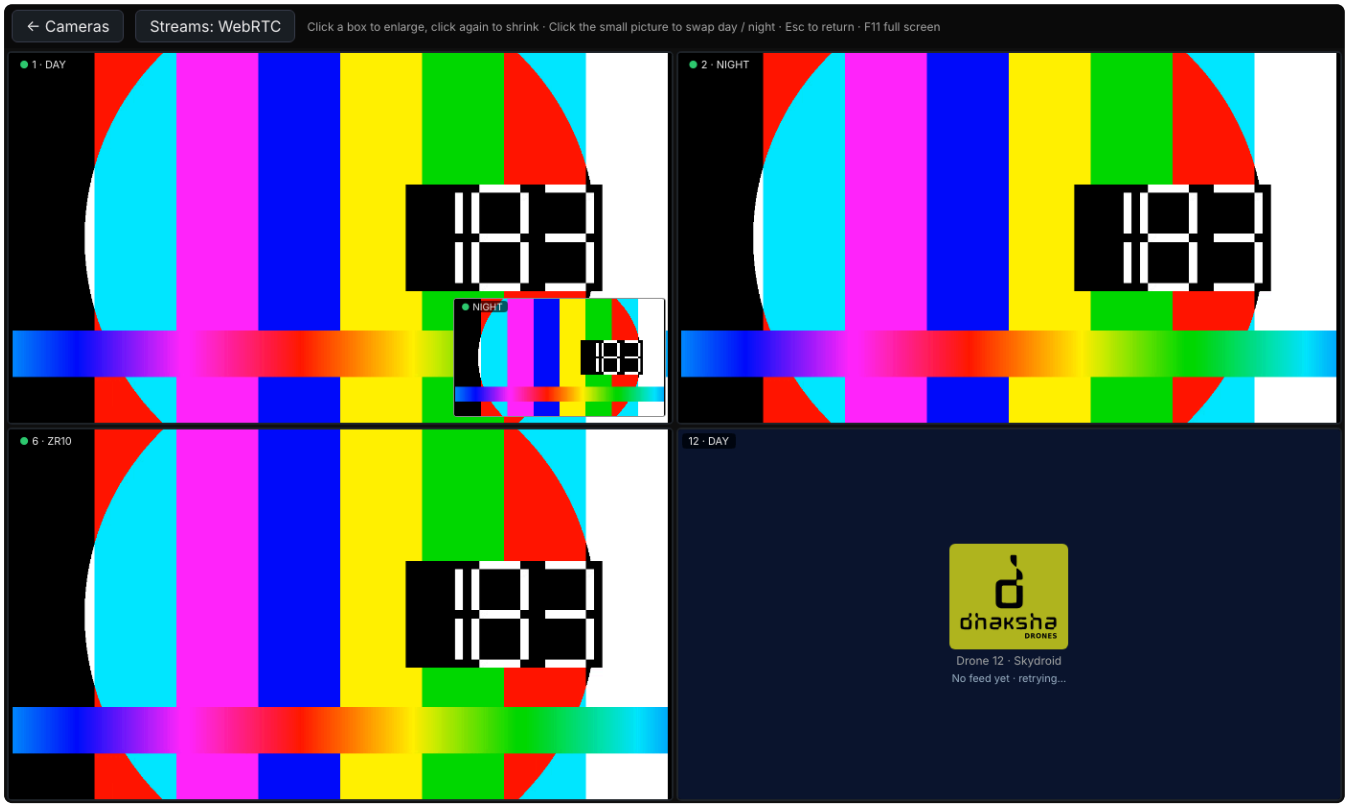
Camera window on a PC. Drones 1-5 Skydroid Day + Night, 6-10 ZR10 Day.



The same window on a phone held upright.

On the video screen:

- Each box shows the Dhaksha logo with "Waiting for feed" until that drone streams.
- **Tap a box** to enlarge it to the whole screen; tap again to shrink it.
- With Day + Night, **tap the small corner picture** to swap day and night.
- **Streams: HLS / WebRTC** button: HLS is steadier (a few seconds delay); WebRTC is faster (about 1 second) but some networks block it.
- A red **"Internet speed very slow for receiving Data"** banner, with the drone numbers, means this viewer's internet is too slow for the video. It disappears by itself when the speed recovers. Showing fewer drones or using a better connection helps.
- **← Cameras** (or Back on a phone, Esc on a PC) returns to the camera window. F11 gives full screen on a PC.



Video screen (test picture) with the red slow-internet banner. Box 1 shows Day with Night in the corner; box 2 Night only; box 6 a ZR10; box 12 is waiting for its drone.

6. DhakshaKey step by step

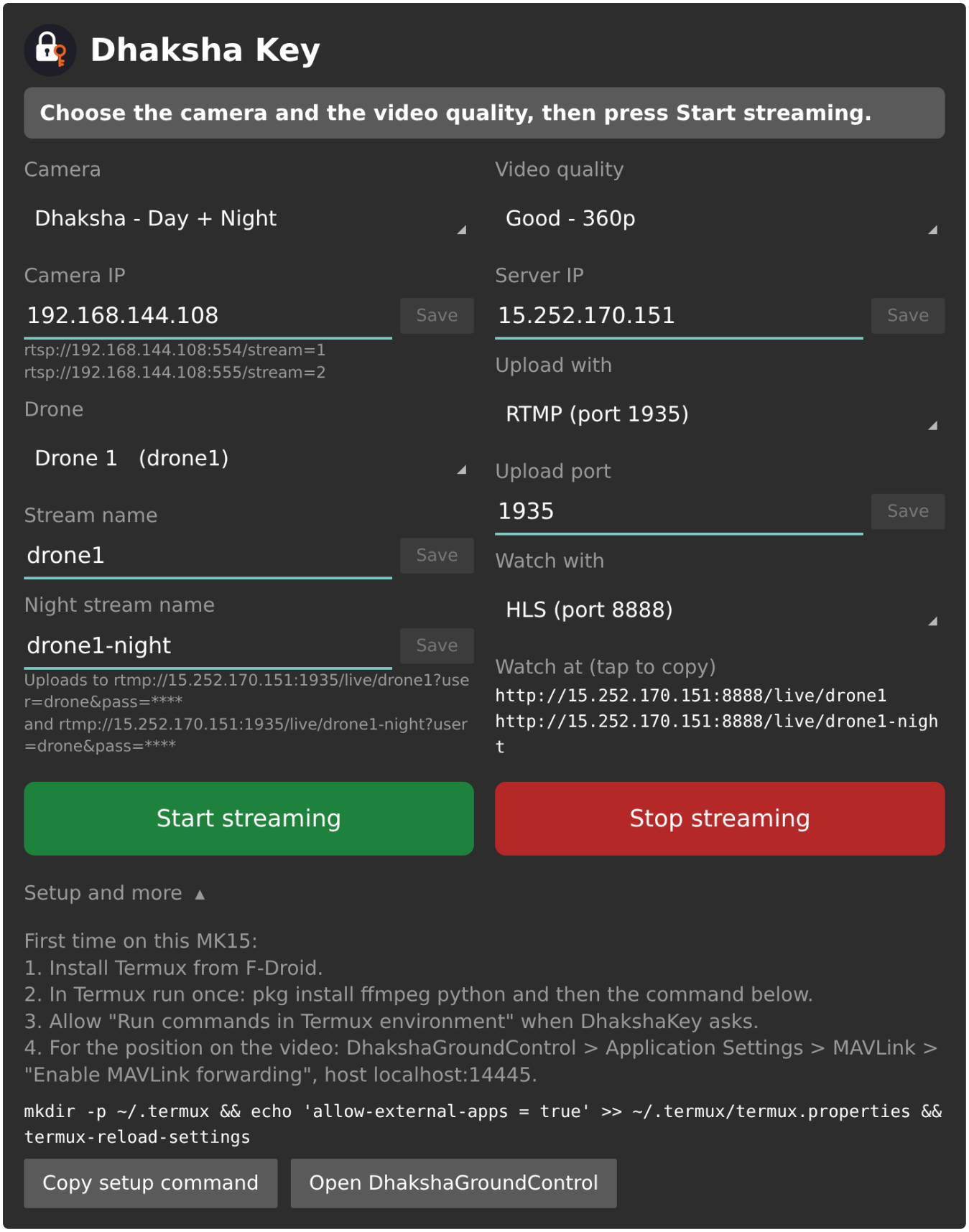
DhakshaKey (version 4.2) is the small app on the controller that sends the drone's live video to the Dhaksha video server. It does the work through a helper app called **Termux**, which opens its own black window while streaming.

6.1 First-time setup (once per controller)

1. Install **Termux** from F-Droid (not from the Play Store).
2. Open Termux and type, then press Enter: `pkg install ffmpeg python` Answer **Y** if asked. Wait until it finishes.
3. Open **DhakshaKey**, tap **Setup and more** at the bottom, tap **Copy setup command**, go back to Termux, paste it and press Enter.
4. Back in DhakshaKey, when Android asks to allow **Run commands in Termux environment**, tap **Allow**.
5. In DhakshaGroundControl, turn on **MAVLink forwarding** to **localhost:14445** (section 4, Setting 2). This puts the drone's position on the video.

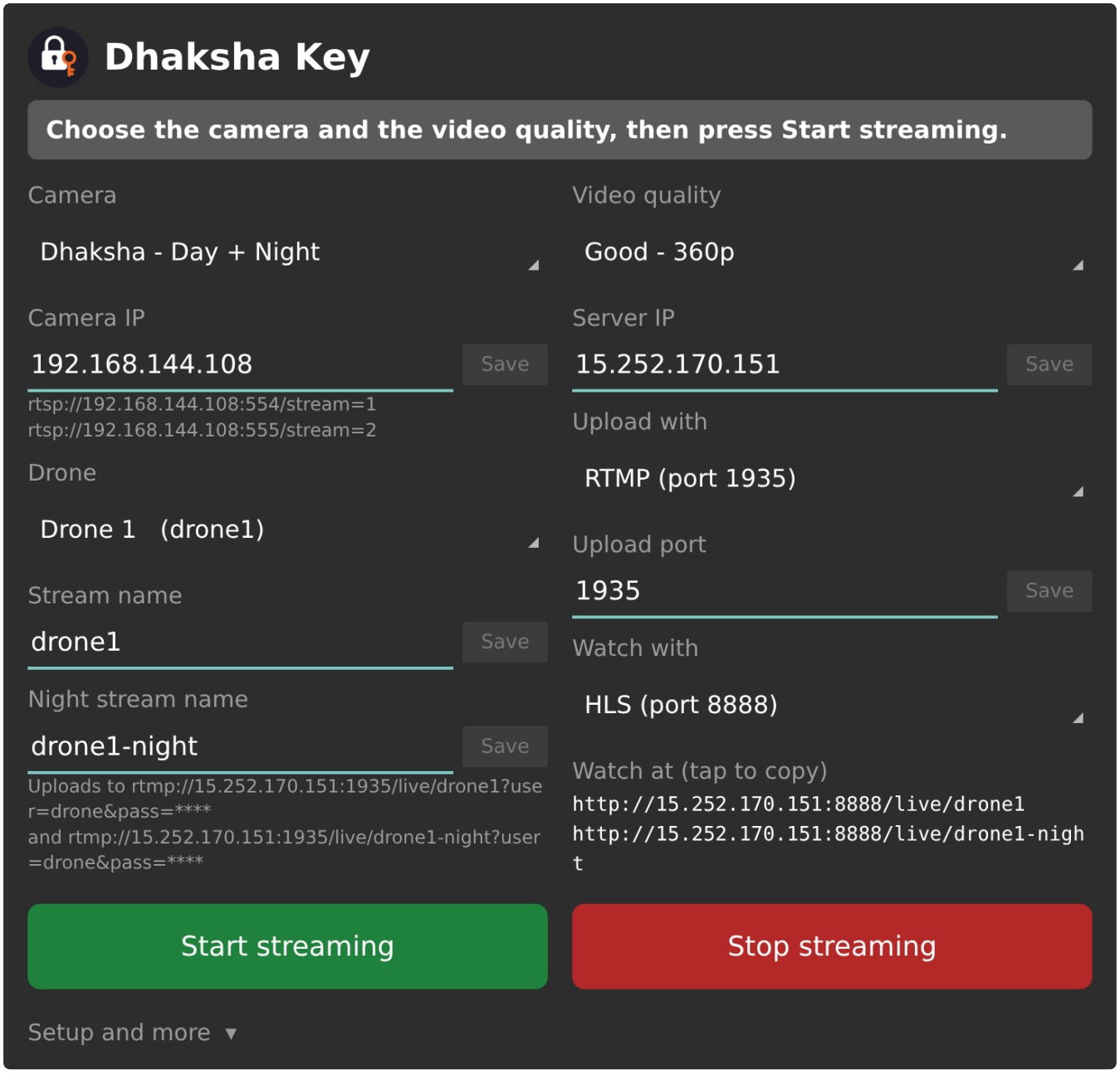
The setup command is:

```
mkdir -p ~/.termux && echo 'allow-external-apps = true' >>
~/.termux/termux.properties && termux-reload-settings
```



Setup and more, opened: the first-time steps, the setup command, and the Copy setup command button. (Drawn from the app's layout, not a photo of the controller; the real screen may differ slightly in fonts and colours.)

6.2 The DhakshaKey screen



DhakshaKey 4.2 ready to start, with Dhaksha - Day + Night chosen (so Night stream name is shown). Left: camera, drone and stream names. Right: quality, server and upload. (Drawn from the app's layout, not a photo of the controller.)

Field	What it is	What to do
Message bar (top)	Coloured bar showing what is happening	Read it after pressing Start
Camera	Which camera to send	Dhaksha - Day, Night, Day + Night, or ZR10

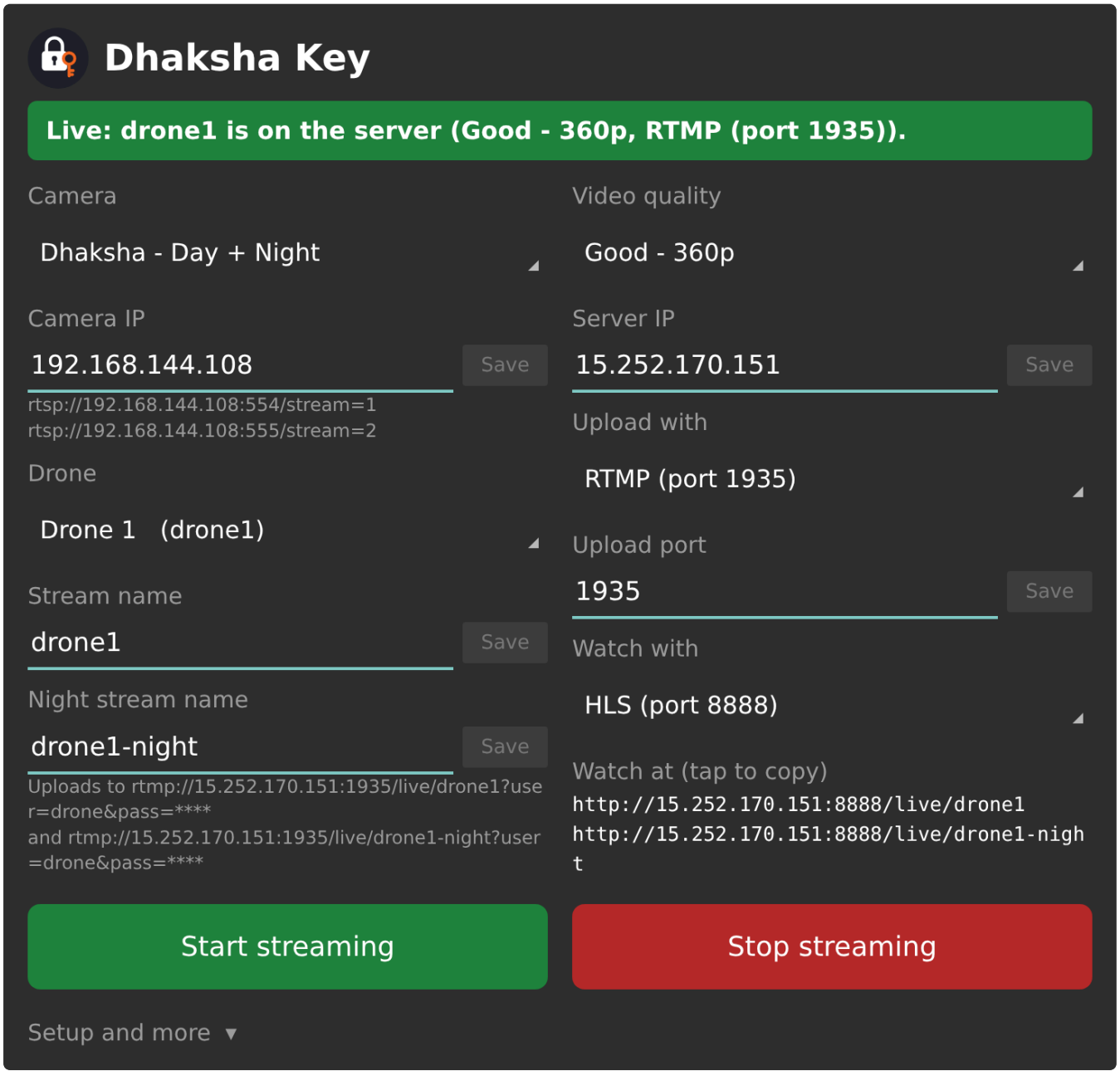
Field	What it is	What to do
Camera IP	Address of the camera on the drone	Normally filled in; change only if told, then tap Save
Drone	This drone's number, 1 to 25	Must be different for every drone in the air
Stream name	Name of the video on the server (drone1…)	Keep the automatic name
Night stream name	Only for Day + Night (drone1-night…)	Keep the automatic name
Uploads to	The full upload address (password hidden)	For checking only
Video quality	Picture size and data use (table in section 5.1)	Start with Good - 360p
Server IP	The Dhaksha video server, 15.252.170.151	Change only if told, then tap Save
Upload with	RTMP, RTSP, SRT or WebRTC (table in section 5.2)	Leave on RTMP
Upload port	Port for the chosen upload type	Leave as filled in
Watch with	HLS or WebRTC for the link below	Either
Watch at	Link to watch this drone in a browser	Tap to copy and share
Start / Stop streaming	Big green and red buttons	See 6.3 and 6.5

The small **Save** buttons stay grey until you change the field next to them. Pressing Start also saves what is typed.

6.3 Every flight: start streaming

1. Power on the drone and connect DhakshaGroundControl as usual; check the camera picture is showing.
2. Make sure the controller has internet (SIM data or Wi-Fi).
3. Open DhakshaKey and choose **Camera**, **Drone** number and **Video quality**.
4. Tap **Start streaming** (green).

5. DhakshaKey first checks the camera answers. If it does, a Termux window opens and the bar shows **Starting drone1 at Good - 360p with RTMP...**
6. Within about 10 seconds the bar turns green: **Live: drone1 is on the server**.
7. Switch back to DhakshaGroundControl to fly. Streaming continues in the background.



Streaming: the bar turns green with "Live: drone1 is on the server". (Drawn from the app's layout, not a photo of the controller.)

If the bar says **The camera is not answering**: check the drone is on and the Camera IP is right, then tap Start streaming again. If the camera is fine but slow to start, tap the orange **Start anyway (camera not answering yet)**.

6.4 While flying: what the bar means

The messages and what to do are in the table in section 5.1. In short: **green** = live; **orange** = short break, it reconnects by itself; **red** = action needed (lower the quality, check the camera, or check the internet). DhakshaKey restarts the stream by itself when the camera or the internet drops, so you do not need to press Start again.

6.5 After the flight: stop streaming

1. Open DhakshaKey.
2. Tap **Stop streaming** (red). The bar shows **Streaming stopped**.
3. You can close the Termux window. (In Termux, **Ctrl+C** also stops it.)

Always stop streaming after landing, so the controller does not keep using mobile data.

6.6 Testing at home without a drone

1. Copy a video file named **test.mp4** into the controller's **Download** folder.
2. In Termux, run once: `termux-setup-storage` and allow access to files.
3. Download the home test script once: `curl -fsSL -o ~/stream_aws_home.sh https://raw.githubusercontent.com/mariappanchellam/qgroundcontrol-v4.2.3/build/dhaksha-live-grid/tools/dhaksha/termux/stream_aws_home.sh`
4. Open it with `nano ~/stream_aws_home.sh` and replace **YOUR_SERVER_IP** with 15.252.170.151 and **YOUR_SERVER_PASSWORD** with the server password (ask your supervisor). Save with Ctrl+O, Enter, Ctrl+X.
5. Run: `bash ~/stream_aws_home.sh`
6. The test video is sent as drone1 with a made-up TEST position. Watch it in Dhaksha Live. Stop with Ctrl+C.

The position text needs DhakshaKey to have streamed at least once on that controller (it installs the position reader). Without it the test video is still sent, just without the position.

7. Quick problem solving

Problem	Check first	Then
No camera picture in DhakshaGroundControl	Drone powered on and radio linked	Settings > Gimbal Camera > Restore Defaults > Apply and Start Video

Problem	Check first	Then
Status says "no reply from camera"	Camera Vendor matches the fitted camera	Check Camera IP; tap Apply and Start Video
Camera buttons do nothing	Connection dot is green	Restart the drone camera, then tap Apply and Start Video
No Day / Night button or Night palette	Camera Vendor is Skydroid	Night RTSP URL is filled in (Restore Defaults)
Night palette does not change the picture	Main picture is on the night camera	Try another palette; report which ones do not work
ZR10 picture freezes	Video = Smooth	Try Reliable (TCP)
"No SD card" / "SD card full"	Card in the camera	Insert or empty the card before flying
DhakshaKey: Data rate too low	Mobile signal	Lower Video quality; use Day only instead of Day + Night
DhakshaKey: Cannot reach the server	SIM data or Wi-Fi on	Server IP 15.252.170.151, Upload port matches Upload with
Dhaksha Live box stays on "Waiting for feed"	DhakshaKey shows Live for that drone	Same drone number and stream name; Server IP and Port in the camera window
Dhaksha Live: red slow-internet banner	Viewer's own internet	Show fewer drones; switch Streams to HLS; use a faster connection
Dhaksha Live in WebRTC shows "blocked"	Network allows WebRTC	Switch Streams to HLS

When reporting a problem, note: drone number, camera type, the exact message on screen, and the time.